

THEJAS HARIDAS

AI/ML Engineer | Data Scientist

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building production LLM pipelines, biomedical knowledge graphs, and document intelligence systems. Specialises in LLM orchestration (vLLM, LangChain, LangGraph), agentic workflows, and graph-based knowledge representation with Neo4j. Adept at engineering robust extraction pipelines with self-correction loops, retry logic, and structured prompt systems for regulated, data-intensive domains.

EXPERIENCE

Junior AI/ML Engineer

Sep 2025 – Present

FeatherSoft, Kochi, Kerala (Full-time)

- Built a multi-stage LLM pipeline to extract structured data from clinical toxicology PDFs using a locally hosted vision model (Qwen via vLLM), processing rasterised document pages into structured JSON and Excel outputs.
- Designed and maintained a biomedical knowledge graph in Neo4j using the Biolink ontology, enabling semantic querying across preclinical research entities with full-text Lucene search and input sanitisation.
- Engineered structured XML prompt systems (extraction, structure detection, row-fill, Excel code generation) with self-correction loops, retry logic, and layered JSON repair fallbacks to handle model inconsistencies at scale.
- Developed Python tooling for automated report generation from multi-report PDFs with page-offset tracking, sex-split appendix handling, and openpyxl-based Excel output.

Trainee Data Scientist

Jan 2025 – Jul 2025

Tyloones Software Private Limited, Noida (Apprenticeship)

- Built LangChain-based data pipelines integrating SQL Server backends for structured data retrieval and downstream ML workflows.
- Developed time series forecasting models and FastAPI microservices to expose ML predictions via REST endpoints.
- Worked on data engineering tasks including ingestion, transformation, and validation pipelines for business analytics.

Research and Development Intern

May 2024 – Dec 2024

Digital University Kerala, Thiruvananthapuram (Internship)

- Developed OCR-based document processing workflows using Label Studio for annotation and Python for post-processing.
- Contributed to ML research involving image processing and optical character recognition for regional language documents.

SKILLS

Programming & Libraries: Python, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, FastAPI, LangChain, LangGraph, Deep Agents, openpyxl, SQL, PySpark, OpenCV

AI / LLM: vLLM, Qwen Vision Models, Prompt Engineering (XML / CoT), RAG, AI Agents, Agentic Workflows, Self-Correction Loops

Databases & Graph: Neo4j, Biolink ontology, Lucene full-text search, Microsoft SQL Server, ChromaDB, Vector Stores

Infrastructure & Tools: Docker, FastAPI, Git, Label Studio, Streamlit, Matplotlib, Power BI

Soft Skills: Problem-solving, Analytical Thinking, Collaboration, Continuous Learning

PROJECTS

1. Clinical PDF Extraction Pipeline (vLLM + Qwen vision model)

End-to-end pipeline that rasterises clinical toxicology PDF pages, sends them to a locally hosted Qwen vision model via vLLM, and extracts structured hematology/TK tables into JSON and Excel. Features multi-phase extraction (table detection → structure consensus → row fill), self-correction loops, and layered JSON repair fallbacks.

Pipeline: Clinical PDF → Rasterise pages → Qwen VL via vLLM → Table detection → Structure consensus → Row fill → JSON repair → Validation → JSON + Excel

2. Biomedical Knowledge Graph (Neo4j + Biolink ontology)

Knowledge graph system for preclinical research data using Neo4j and the Biolink ontology. Implemented full-text Lucene indexing with input sanitisation, YAML-based configuration, and a Python query interface for semantic entity retrieval.

Pipeline: Preclinical data → YAML config → Biolink mapping → Neo4j graph → Lucene index → Input sanitisation → Python query API → Semantic retrieval

3. Alcohol Sales Prediction (FastText + BiGRU + Attention)

Text classification pipeline using FastText embeddings, BiGRU layers, and a self-attention mechanism for sales forecasting. Included SHAP-based explainability analysis for model interpretability in a regulated context.

Pipeline: Sales text data → Preprocessing → FastText → BiGRU layers → Self-attention → Dense head → Prediction → SHAP analysis

EDUCATION

M.Sc. Computer Science (Data Analytics)	2023 – 2025
Kerala University of Digital Sciences, Innovation and Technology (Digital University Kerala)	
B.Sc. Physics	Graduated 2023
Kerala University	

PUBLICATIONS

- Thejas Haridas et al., “Chaos-based Audio Encryption: A Comparative Study,” *Franklin Open (Elsevier)*, September 2024.

CERTIFICATIONS

- Neo4j Fundamentals – Neo4j (Sep 2024)
- Google Cloud Career Launchpad — Data Analytics Track – Google Cloud (Jun 2024)

ACHIEVEMENTS

- First author on a peer-reviewed publication (Franklin Open, Elsevier) at postgraduate level.
- Qualified UGC NET in Computer Science.